

Scratch: Sequencing & Events

30-Minute Tiered Homework — Due: Thursday July 2

🕒 30 MIN 📁 4 PARTS ALL TIERS 18 PTS + 3 BONUS

✓ TEACHER ANSWER SHEET — SESSION 3 HOMEWORK

👤 STUDENT NAME

Teacher Copy — Do Not Distribute

📅 DATE

July 2, 2026

📖 GRADE / TIER

All Tiers

🕒 **PLAN: 5 min** Part 1 — Predict & Bug Fix · **10 min** Part 2 — Scratch Animation · **10 min** Part 3 — Written Challenge · **5 min** Part 4 — Reflect & Share

PART 1 OF 4 — 5 MIN, NO SCREEN

Predict the Output & Fix the Bug

🔗 PUZZLE A — PREDICT WHAT HAPPENS

The blanks are intentional — students fill them in. Model answers shown in green.

```
when green flag clicked
move [blank] steps
say "Hello!" for [blank] secs
move [blank] steps
say "Bye!" for [blank] secs
```

📖 Blanks can be any number — accept any reasonable values. The logic flow is what matters, not the specific numbers. Example with 10 steps / 2 secs:

What triggers the script?

✓ Event (Hat Block): **when green flag clicked** — the green flag is the trigger event that starts this script.

Describe the sprite's actions in order:

1. Moves right X steps
2. Says "Hello!" for Y secs
3. Moves right X more steps
4. Says "Bye!" for Z secs

Key insight: Steps execute in order. The sprite finishes each step completely before starting the next one (no multitasking within a single script).

🔗 PUZZLE B — SPOT THE BUG 🐛

```
move 10 steps ← bug here?
play sound "pop"
when [space] key pressed
```

💡 Hint: look at where the hat block is!

What is the bug?

✓ The **when [space] key pressed** hat block is at the **BOTTOM** of the stack instead of the **TOP**. The **move** and **play sound** blocks are floating — they have no hat block above them, so they **NEVER** run.

How would you fix it?

✓ Move the hat block to the **TOP**. The corrected script order is:

```
when [space] key pressed ← HAT BLOCK
FIRST ✓
move 10 steps
play sound "pop"
```

📖 Rule: EVERY script starts with a hat block at the very top. This is the #1 beginner Scratch bug. If a script never runs — check for a missing or misplaced hat block!

PART 2 OF 4 — 10 MIN, ALL TIERS

Finish Your Scratch Animation

🕒 TASK — ALL TIERS (SAME GOAL, DIFFERENT ON-RAMPS)

Open your Scratch project. Make sure your sprite does **at least 3 actions in order** when the green flag is clicked.

🔗 SHARE YOUR PROJECT

In Scratch: **Share** → **Copy Link**. Paste below, or attach a screenshot.

My Scratch project link:

✓ Accept any valid [https://scratch.mit.edu/projects/\[ID\]](https://scratch.mit.edu/projects/[ID]) URL. If student has no internet, accept a screenshot showing the project running.

- ✓ **TIER A** — 3 blocks on green flag; sprite moves AND says two different things.
 - ✓ e.g. move 10 → say “Hi!” → say “I moved!” — Award if sprite visually responds to flag click with 3 distinct actions.
- ✓ **TIER B** — Script A: 3 actions on green flag. Script B: 1 action on key press.
 - ✓ Two separate scripts with two different hat blocks. Award if both scripts run independently and project is named.
- ✓ **TIER C** — Both scripts above + a second sprite with its own event & 3-action sequence.
 - ✓ Award if 2 sprites each have independent event-driven scripts. Bonus if sprites interact or respond to each other.
- ✓ **Project is named & saved.**

Award 1 pt for naming. Deduct if project is untitled or not saved to Scratch cloud.

Project name:

- ✓ Any descriptive name (not “Untitled”). Reward creative names!

Describe what your sprite does (2-3 sentences):

- ✓ **Sample:** “When I click the green flag, my cat sprite moves 10 steps to the right and says ‘Hello!’ Then when I press the space bar, it jumps up and says ‘Whee!’” Award full marks for any coherent 2-3 sentence description that matches their Scratch project.

 **Grading tip:** Compare the written description to the project if possible. Reward accuracy and specificity. “It moves” = partial. “It moves 10 steps to the right then says ‘Bye!’ for 2 seconds” = full.

Tiered Written Challenge

Circle your tier: **Tier A** **Tier B** **Tier C**☆ **TIER A — FOUNDATIONS**

Draw the blocks from your project. Label each block with its colour and name.

Drawings should show blocks in the correct Scratch colours: yellow=Events, blue=Motion, purple=Looks. Each block must have a label.

1. What event starts your script?

✓ Event: **when green flag clicked** — the yellow hat block is the trigger.

2. What does your sprite do first? Second? Third?

✓ First: moves (Motion block). Second: says something (Looks block). Third: any 3rd action described in correct order. Award if sequence is logical and matches their project.

3. One thing you want to add tomorrow?

✓ Open-ended. Any valid Scratch addition counts: "I want to add a costume change" / "I want it to play a sound" / "I want to add another sprite."

💧 **TIER B — CORE**

Fill in the table for your two scripts.

SCRIPT	EVENT (HAT BLOCK)	ACTIONS IN ORDER
A	e.g. when green flag clicked	move → say "Hi" → change costume
B	e.g. when [space] key pressed	say "Jump!" or change y by 20

Accept any 2 different hat blocks. Key check: Script A and B must have DIFFERENT events.

Can Scripts A and B run at the same time? How?

✓ Yes — IF the player presses space at the same moment they click the green flag. Scratch can run multiple scripts simultaneously if their events fire at the same time. This is called parallel execution.

1. Difference between a script and a block?

✓ A script is the full stack of connected blocks (the complete program). A block is a single instruction within the script (one step). Script : block = recipe : ingredient.

2. What if you drag a block away from the hat block?

✓ It will NEVER run automatically — it becomes a disconnected, "floating" block. Scratch only runs blocks that are connected under a hat block.

3. ONE change from class → home. What did you notice?

✓ Open-ended. Award for honest reflection: "I changed the number of steps — the sprite moved farther." / "I changed the event to arrow key — now I can control it!"

✂ **TIER C — ACCELERATED**

Design Challenge: Plan a 3-sprite project for Sessions 4-5.

SPRITE	EVENT	3 ACTIONS
1	when green flag clicked	move → say "Go!" → change costume
2	when [right arrow] pressed	change x by 10 → play sound → say "Moving!"
3	when [space] pressed	change y by 20 → wait 0.5s → change y by -20

Theme of your 3-sprite project?

✓ Accept any creative theme: Space Race, Animal Farm, Sports Game, Adventure, etc.

🤖 **AI Connection — Model Answers**

Both run? I predict...

✓ Yes — if both start with **when green flag clicked**, BOTH scripts fire simultaneously. Scratch runs all matching event scripts in parallel. They are independent — one doesn't wait for the other.

Similarity to AI?

✓ "Both follow precise, ordered instructions. AI uses algorithms (like a Scratch script) to process inputs and produce outputs — step by step, in sequence. An AI's 'event' is receiving new data; its 'script' is the trained model running its decision steps."

3-Sprite Scratch Project translated to Python pseudocode

```
def when_green_flag_clicked(): # Sprite 1's script
    sprite1.move(10)
    sprite1.say("Go!", duration=2)
    sprite1.next_costume()
```

```
def when_right_arrow_pressed(): # Sprite 2's script
    sprite2.x += 10
    sprite2.play_sound("pop")
    sprite2.say("Moving!", duration=1)
```

AI Connection: AI also follows ordered steps when processing input

Input (event) → Process (script) → Output (sprite action)

✓ Award bonus (+3 pts) for valid pseudocode with correct Python function syntax, at least 2 sprites translated, and AI connection comment.

PART 4 OF 4 — 5 MIN

Reflection & Family Share-Out

🗣️ QUICK REFLECTION — 1 SENTENCE EACH

The most interesting thing I learned today was:

✓ **"I learned that the ORDER of blocks matters — if I move a block to the bottom it stops working!"** Accept any genuine reflection on sequencing, events, or Scratch.

One question I still have:

✓ **Open-ended. Award for any curious question: "Can two sprites talk to each other?" / "What happens with 10 sprites?" / "Can I make a game with these blocks?"**

Something that was hard but I kept trying:

✓ **Growth mindset reflection. Award for honesty + persistence: "Getting the blocks in the right order was hard but I tried moving them around until it worked."**

I am proud of:

✓ **Any achievement: "I'm proud that my sprite moved AND said something when I pressed the key — I figured out the bug myself!"**

🏠 FAMILY SHARE-OUT

Explain one thing you learned to a parent, guardian, or sibling. Use these words:

event sequence script sprite

What did you explain to them?

✓ **Sample: "I explained that in Scratch, a sprite is the character I program. I put blocks in a script to tell it what to do. The sequence has to be in the right order. An event (like clicking the green flag) starts the whole thing running!"**

Award full marks if all 4 vocabulary words are used correctly in their explanation.

📊 SCORING

PART	POINTS
Part 1 Predict & Bug Fix	4 pts
Part 2 Project Finished + Link	6 pts
Part 3 Tiered Written	5 pts
Part 4 Reflection + Share	3 pts
Total	18 pts
Tier C Bonus (3-sprite + AI)	+3 pts

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